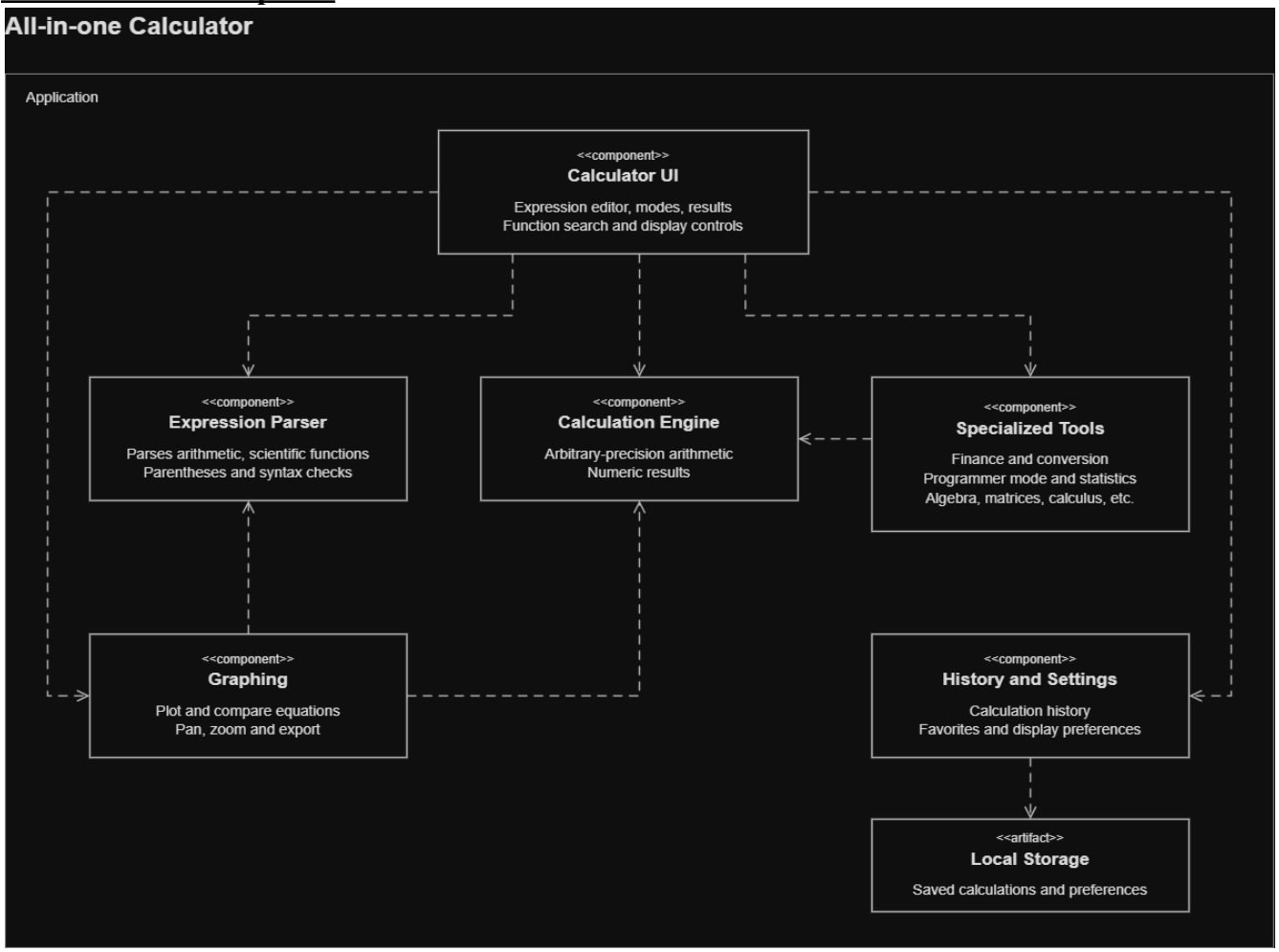
Include Team Number, Team Members, Project Name (1–3 words), Project Synopsis (1–25 words), Architecture Description (500–1000 words, 3 distinct UML diagrams).

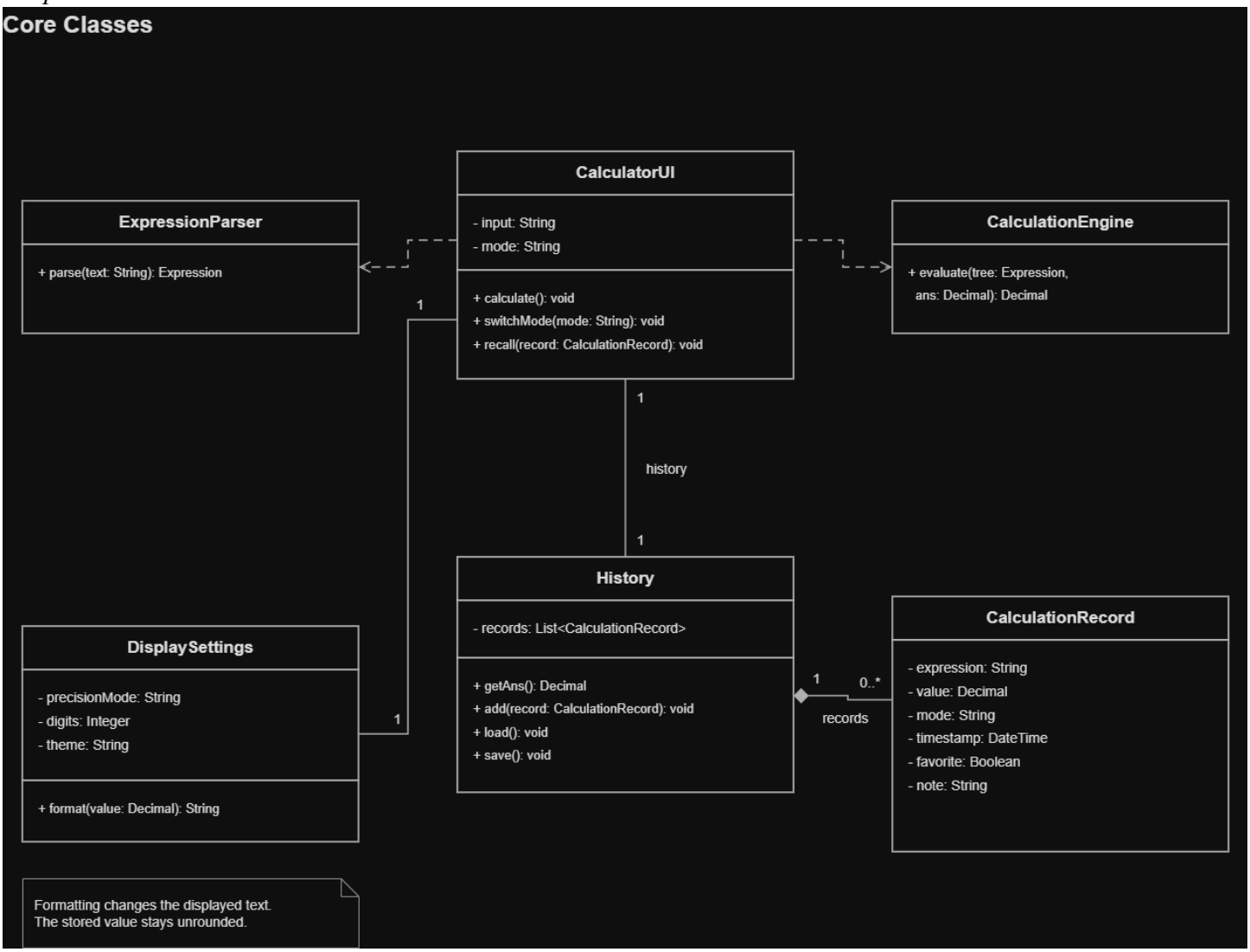
Diagrams should show different software aspects (e.g., data flow, components).

Team Number: 20
Team Members: Ivan Kullaya, Pruthviraj Sadhankar, Liam Kinghouser, Gael Salazar-Morales, Joshua Fakunmoju, Gabriel Haro-Villa, Carter Ruff
Project Name: All-in-one Calculator
Project Synopsis: A calculator that consolidates basic math, scientific calculations, financial tools, unit conversions, and advanced graphing functionality into a single unified platform.
Architecture Description:



Description: An all-in-one calculator app that consolidates basic math, scientific calculations, financial tools, unit conversions, and advanced graphing functionality into a single unified platform. It does not just provide isolated numerical answers; it allows users to plot equations on interactive graphs, track previous calculation history, and instantly switch between different calculation modes based on their exact needs. Different mathematical features where users can filter results by specific categories such as algebra, statistics, finance, or trigonometry to get tailored information for each function. Clicking on each calculation mode displays advanced parameters, step-by-step breakdowns, formula definitions, and customizable variables so users can tweak inputs and instantly see updated outputs. A programmer mode will provide number conversions between decimal, binary, and hexadecimal as well as bitwise operations. The specialized modules will allow these functions to be added or modified without requiring changes to the entire calculator. The financial mode will similarly handle functions such as compound interest and loan payments, while the standard conversion module will handle common length, weight, volume, temperature, and currency conversions. The calculator application will also contain an input validation component, such

as checking an expression before outputting the result to see for invalid input and provide useful or simple explanations when an error of missing parentheses or division by zero occurs. It will also support undo and redo operations so users can easily correct changes while entering an expression, as well as providing a standard 'Ans' function that will allow the users to use the result of a previous calculation in a new expression. The history and saved-calculation component will manage information that users may want to access again, which the history will allow the users to review previous inputs and outputs. The saved-calculation system will allow frequently used calculations to be saved for quick access. It will also allow for users to enter text as short notes for saved calculations. Financial scenarios can be saved separately so that multiple sets of financial results can be compared side by side, providing users with a way to organize calculations instead of repeatedly entering the same information in each expression. The graphing component will allow users to plot and compare equations, and will have graphing controls such as panning, zooming and export. The user interface will connect these modules and provide a consistent way for users to interact with the application. The mode switcher will function as said above, all while preserving their current input and session history. The category filter will allow users to search for functions based on mathematical functions. The interface will also provide calculation breakdowns, formula definitions and adjustable decimal or significant-figure settings. The full stored value can remain available internally even when the displayed result is rounded for readability. Finally, the calculation will support user customization and persistent features. Users will be able to change the system UI color theme, dark or light mode. Saved calculations, favorites, history, notes, and other user information can be retained for later access. Calculation history will be exportable as a text or CSV file. The system will remain organized by separating the application into calculation, graphing, financial, conversion, history, and interface components.



Sequence Diagram

